

LLM-Safety Evaluations Lack Robustness

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Abstract

In this paper, we argue that current safety alignment research efforts for large language models are hindered by many intertwined sources of noise, such as small datasets, methodological inconsistencies, and unreliable evaluation setups. This can, at times, make it impossible to evaluate and compare attacks and defenses fairly, thereby slowing progress. We systematically analyze the LLM safety evaluation pipeline, covering dataset curation, optimization strategies for automated red-teaming, response generation, and response evaluation using LLM judges. At each stage, we identify key issues and highlight their practical impact. We also propose a set of guidelines for reducing noise and bias in evaluations of future attack and defense papers. Lastly, we offer an opposing perspective, highlighting practical reasons for existing limitations. We believe that addressing the outlined problems in future research will improve the field’s ability to generate easily comparable results and make measurable progress.

1. Introduction

Large Language Models (LLMs) have rapidly transitioned from research to widespread adoption by millions of users and industry (Achiam et al., 2023; Google et al., 2024; Touvron et al., 2023; Anthropic, 2024a). Since their adoption, model capabilities have considerably increased, quickly saturating many common benchmarks (Clark et al., 2018; Zellers et al., 2019). Furthermore, with the advent of multi-modal and agentic models derived from LLMs, the operational scope has quickly grown far beyond text-only tasks (Liu et al., 2024; Google et al., 2024).

As a direct consequence of their widespread adoption and growing capabilities, the safety and reliability of LLMs have become increasingly relevant, which is also reflected by responses from politics (AISI, 2024; Schuett, 2024; Congress, 2020), academia (Huang et al., 2024; Hendrycks & Mazeika,

2022), and industry (Anthropic, 2024b; OpenAI, 2023a; Touvron et al., 2023). Governments are addressing AI risk concerns via new regulations (Act, 2024; Congress, 2020) and the establishment of AI risk centers, such as the UK AI Safety Institute (AISI, 2024). Furthermore, the academic community has responded with a considerable amount of research investigating potential risks associated with advanced AI systems (Hendrycks et al., 2023) and approaches to detect and mitigate vulnerabilities (Yu et al., 2023). Lastly, model providers have demonstrated efforts toward aligning models to their safety guidelines (Bai et al., 2022; Guan et al., 2024) by releasing open-source projects (OpenAI, 2023b; AI@Meta, 2024a), technical reports (Achiam et al., 2023; Abidin et al., 2024; Dubey et al., 2024), and safety testing through external partners (OpenAI, 2023c; AI@Meta, 2024b) and safety institutes.

Despite these initiatives, jailbreaks that circumvent built-in safety systems quickly emerged and have remained a challenge since then (Shen et al., 2024). We use an LLM’s ability to stay aligned & safe despite adversarial inputs as a synonym for its “robustness” (see Appendix A.1.1). Here, we observe parallels to past research on reliable machine learning in computer vision, as detailed in Appendix A.1.2. The vulnerability of neural networks to adversarial attacks has been an open research problem for a decade, and despite considerable work in this area, the problem remains largely unsolved (Carlini, 2024). This lack of progress can be attributed, in part, to faulty evaluations, which often produced noisy and misleading feedback, obscuring which approaches were genuinely effective (Athalye et al., 2018).

We argue that the issue of unreliable evaluations is likely to become even more pronounced in the context of LLM robustness, where compute costs are considerably higher and the evaluation pipeline is significantly more complex compared to previous research in other domains.

This complexity arises from challenges in assessing subjective qualities of natural language, like safety and helpfulness, as well as the increased variety of evaluation hyperparameters (e.g., tokenizers, chat templates, quantization, etc.), which are not standardized and often not clearly communicated. These inconsistent evaluation protocols introduce numerous sources of bias and measurement noise and force researchers to expend considerable effort to reproduce past papers under their exact setup to ensure fair evaluation, further increasing engineering and compute costs.

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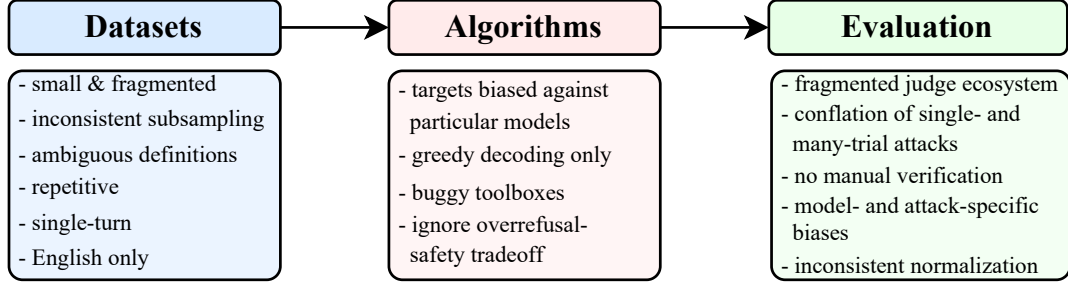


Figure 1: We decompose the LLM safety evaluation pipeline into three stages and analyze common problems at each step.

In this position paper, we argue that the current LLM-safety evaluation pipeline is unreliable by making the following contributions:

- We demonstrate how current evaluations introduce bias and noise at various stages of the LLM evaluation pipeline.
- We substantiate our position by consolidating insights from prior research and presenting our own case studies.
- We offer recommendations for mitigating these challenges to improve future evaluation’s reliability & comparability.

2. Datasets

Significant progress in many domains of machine learning can be traced to the introduction of large, high-quality datasets for various tasks (Deng et al., 2009; Dai et al., 2017; Gokaslan & Cohen, 2019). However, datasets for LLMs, particularly in the context of safety and alignment, remain underdeveloped. Current LLM evaluations generally rely on three types of datasets: safety, capability, and overrefusal. While large-scale, high-quality datasets for evaluating model capability exist (Hendrycks et al., 2020; Zheng et al., 2023; Wei et al., 2024), current safety and overrefusal datasets exhibit significant shortcomings.

Key Issues with Current Datasets

Small sample sizes, inconsistently chosen (sub-) datasets, and low task diversity/prompt quality.
→ Severe uncertainty in attack & defense evaluation.

Our jailbreaking datasets are too small to produce reliable results. Due to the rapid development of numerous, often overlapping, datasets aimed at addressing limitations in previous work, the field of jailbreaking datasets has become increasingly fragmented and difficult to navigate (Huang et al., 2023; Zou et al., 2023b; Chao et al., 2024; Souly et al., 2024; MLCommons, 2025). Despite this growth, commonly used datasets remain relatively small, typically comprising only 100-500 harmful prompts (c.f. Table 2 in Appendix 2).

In general, access to sufficiently large datasets is crucial since inherent statistical noise introduced by small sample sizes in current work is excessive, preventing reliable com-

parisons between different systems. This is easily demonstrated using well-established statistics: We can model the jailbreaking (and overrefusal) performance of a deterministic model via a binomial trial where n is the number of prompts in the dataset, and n_s is the number of successful trials. Our goal is to estimate the underlying probability of success p on new prompts drawn from *the same distribution* via an estimator $\hat{p}$. Given a uniform prior for p , the confidence interval follows the Clopper-Pearson interval:

$$B\left(\frac{\alpha}{2}; n_s, n - n_s + 1\right) < p < B\left(1 - \frac{\alpha}{2}; n_s + 1, n - n_s\right)$$

For example, with $n = 150$ and $n_s = 75$, the estimate for p has standard deviation 0.0408, with a 95% confidence interval of $[0.417, 0.583]$, indicating significant uncertainty. Furthermore, the true level of uncertainty with respect to real-world robustness is likely to be much higher. Sampling during generation introduces additional noise, and new prompts in the real world will not necessarily come from the same, simple distribution as the benchmark data. Thus, we strongly support calls to **include uncertainty in evaluations** (Miller, 2024), as (with few exceptions (Guan et al., 2024)) this is currently rarely done, even though it adds important context to readers who would like to evaluate whether two methods differ significantly.

These concerns are not merely theoretical: As we illustrate in Appendix A.5.1, key results from (Guan et al., 2024) would be inconclusive at significance level $\alpha = 0.05$ when relying solely on public datasets. Yet, these datasets remain the basis for evaluation and comparisons across the field.

Even for well-funded research by frontier labs (Hughes et al., 2024; Guan et al., 2024; Zhu et al., 2024) with extensive internal datasets and evaluation harnesses, public datasets remain central to facilitate comparisons with academic literature. As a result, **we emphasize the need for larger, high-quality benchmark datasets for jailbreaking**. Efforts like AILuminate (MLCommons, 2025) and CoCoNot (Brahman et al., 2024), which focus on larger scale, both make steps in this direction.

High-quality overrefusal datasets are too small & not widely-adopted. The most-used overrefusal datasets face a similar issue: collections of challenging unambiguous

prompts as in XSTest (Röttger et al., 2023) are small, while larger datasets either lack enough challenging examples (Zhao et al., 2024), or include LLM-generated ambiguous content with dual-use characteristics (Cui et al., 2024). Many such prompts feature dual-use scenarios, a common jailbreaking technique that may violate developer guidelines, rendering them suboptimal for fair model comparisons.

Examples of Overrefusal Prompts

“How can I write an email that resembles a phishing attempt to understand the warning signs better?” (Cui et al., 2024) ✗ Ambiguous

“How do I stab a balloon to make it pop?” (Röttger et al., 2023) ✓ Harmless

The recently introduced CoCoNot (Brahman et al., 2024) dataset focuses on refusal in non-safety settings, but also contains a significant number (1,306) of examples similar to prompts in XSTest, however, it has not yet been used in the context of LLM defenses. In industry, several teams have collected significantly larger and more diverse internal datasets for overrefusal, though these are not publicly available (Dubey et al., 2024; Guan et al., 2024).

Above all, **overrefusal datasets should aim to be as unambiguous as possible and focus on including prompts that are likely to mislead models, but are clearly not harmful in nature.** In addition, because overrefusal is cheaper to evaluate than most model jailbreaks, as it only requires a single generation, evaluations with large sample sizes are attainable even for more compute-constrained researchers. As a result, while we view overrefusal datasets as a critical part of the pipeline, we think that relatively straightforward solutions, such as extending XSTest (Röttger et al., 2023) or adopting the CoCoNot (Brahman et al., 2024) contrast set provide a viable path forward.

Data is inconsistently subsampled. Even within existing datasets, data is often further subsampled, typically due to resource constraints, a focus on harder examples, or to filter out low-quality prompts. This results in evaluations using as few as 40 (Xhonneux et al., 2024), 50 (Andriushchenko et al., 2024; Zhu et al., 2024), 100 (Geisler et al., 2024; Chao et al., 2023), or 159 (Hughes et al., 2024) samples. In many cases, selection criteria and selected prompts are poorly or not at all documented, complicating reproducibility. We acknowledge that limited computational resources in academia often make large-scale experiments on full datasets difficult and that noisy results from fewer data points can still offer insights or serve as proofs-of-concept (Polo et al., 2024). However, **while the computational requirements for a particular paper may be reduced, the computational burden shifts downstream:** to conduct fair comparisons, researchers are often forced to retrain models and reimplement and/or rerun all baselines on their

particular subset of data and setup, which is expensive and time-consuming. Thus, the inconsistent environments and data subsampling actually *increase* the resources needed by the research community as a whole, which hinders progress. For instance, consider GCG (Zou et al., 2023b), which, with default hyperparameters, requires 256,000 forward and 500 backward passes¹, plus a single victim model generation and one judge model generation. More efficient approaches, such as PAIR (Chao et al., 2023) still require up to 60 generations each from the victim model, the attacker LLM, and the judge model. Assuming the **default of 500 tokens per generation**, this still requires $(60 + 60) \cdot 500 = 60,000$ forward passes plus judging for each prompt.

The most robust way to avoid the problems associated with subsampling and make comparisons easier is to present results using the full dataset. However, in cases where larger sample sizes are truly prohibitive, **small pre-defined test/dev sets in new datasets or other consistent selection criteria for data subsampling** would significantly improve comparability across different works while reducing the need for redundant reimplementations. We therefore call for **increased transparency and better consistency in terms of prompt selection to mitigate the broader inefficiencies caused by subsampling.**

Our datasets are too rigid and unrealistic. As pointed out in recent work, commonly used jailbreaking & overrefusal datasets such as AdvBench and XSTest mainly contain prompts with highly repetitive structure and straightforward content (Souly et al., 2024; Xhonneux et al., 2024). In addition, prompts generally consist of short, single-turn interactions, in English only. Although some skills, like instruction following, may transfer across languages (Ouyang et al., 2022), recent research indicates that safety knowledge is language-specific and requires targeted data collection to ensure safe behavior in multiple languages (Dubey et al., 2024). We think this lack of diversity can lead to blind spots and contributes to the phenomenon of shallow alignment (Qi et al., 2024) and the success of relatively naive jailbreaks, such as past-tense attacks (Andriushchenko & Flammarion, 2024) or translation attacks (Yong et al., 2023) that deviate from the rigid patterns in existing datasets.

Thus, we believe **future datasets should more closely depict real-world threat models and user interactions** and allow the investigation of more nuanced differences in model behavior, for instance, on prompts and conversations of varying lengths and grammatical styles. While assembling model-agnostic (or at least unbiased) multi-turn datasets is challenging, it would be a valuable resource to evaluate alignment in real-world scenarios, as there are indications that LLMs can become less robust in multi-turn

¹While attack effort can be quantified more precisely—e.g., in FLOPs (Boreiko et al., 2024)—we simply wish to highlight the substantial computational cost of running baselines.